Kitāb al-Mukhtaṣar fï Ḥisāb al-Jabr wa-l-Muqābala

كتاب المختصر في حساب الجبر والمقابلة

The Compendious Book on Calculation by Completion and Balancing

And on the laws of inheritance (al-farā'iḍ) and the distribution of legacies

Muḥammad ibn Mūsā al-Khwārizmï

محمد بن موسى الخوارزمي

Critical Edition

'Alī Muṣṭafā Musharrafa and Muḥammad Mursï Aḥmad
D. Phil. (London) & D. Phil. (Edinburgh)
Professors of Mathematics, Egyptian University

Prepared under the auspices of the Faculty of Science, Egyptian University
With the support of the Egyptian Association for the Advancement of Science

Cairo: Maṭba'at Fatḥ Allāh Ilyās Nūrï wa-Awlāduh
1359 AH / 1939 CE

Contents

Editors’ Preface

تعنى الأمم بتراثها العلمي لأنه نوع من الغذاء الروحي لعلمائها ومفكريها وسائر المتعلمين فيها. ولعلنا نحن المصريين أغنى الأمم تراثاً فقد تعاقبت علينا حضارات مختلفة منذ فجر التاريخ إلى اليوم، وفي كل حضارة منها قمنا بقسط وافر من واجبنا العلمي نحو الأسرة البشرية. وليس يكفي أن نتحدث عن مجدنا العلمي كما لو كان أسطورة أو حديث خرافة، يتغنى به الشعراء ويتعالى في وصفه الخيال، بل يجب أن يظهر هذا المجد في صورة ملموسة تراها الأعين وتنالها الأيدي. لذلك كان من المهم أن نعنى بنشر الكتب التي وضعها آباؤنا وأجدادنا خصوصاً إذا كانت هذه الكتب هامة الأثر في تكييف التفكير البشري. ولا شك أن في مقدمة هذه الكتب كتاب الخوارزمي في الجبر والمقابلة.

Nations concern themselves with their scientific heritage because it is a kind of spiritual sustenance for their scholars, thinkers, and all their students. We Egyptians may well be the richest of nations in heritage, for diverse civilizations have succeeded one another upon our soil from the dawn of history to the present day, and in each of them we have made a generous contribution to our scientific duty toward humanity.

It is not sufficient to speak of our scientific glory as though it were a legend or a fairy tale, sung by poets and elevated by imagination; rather, this glory must appear in a tangible form, visible to the eyes and graspable by the hands. Therefore it was important to concern ourselves with the publication of books written by our ancestors, especially those works of paramount influence in shaping human thought. Undoubtedly foremost among these is the book of al-Khwārizmī on Algebra and Almucabala.

The Author's Preface

In the Name of God, gracious and merciful! Praised be God for His bounty towards those who deserve it by their virtuous acts: in performing which, as by Him prescribed to His adoring creatures, we express our thanks, and render ourselves worthy of the continuance (of His mercy), and preserve ourselves from change: acknowledging His might, bending before His power, and revering His greatness! He sent Mohammed (on whom may the blessing of God repose!) with the mission of a prophet, long after any messenger from above had appeared, when justice had fallen into neglect, and when the true way of life was sought for in vain.

بسم الله الرحمن الرحيم. الحمد لله على نعمه لأهل طاعته، بأدائها بقدرته على عباده المؤمنين به، ليثيبهم بها ثواباً، ويحفظهم بها عن التغيير، ويعرفهم قدرته، ويذللهم لعظمته. أرسل محمداً صلى الله عليه وسلم برسالته بعد انقطاع الرسل، وذهاب العدل، وفساد النُّصُل.

The learned in times which have passed away, and among nations which have c eased to exist, were constantly employed in writing books on the several departments of science and on the various branches of knowledge, bearing in mind those that were to come after them, and hoping for a reward proportionate to their ability, and trusting that their endeavours would meet with acknowledgment, attention, and remembrance—content as they were even with a small degree of praise; small, if compared with the pains which they had undergone, and the difficulties which they had encountered in revealing the secrets and obscurities of science.

وقد كان العلماء في الأزمان الخالية وفي الأمم السالفة يؤلفون الكتب في أبواب العلوم وفروع المعرفة، يتذكرون من بعدهم، ويرجون الأجر بقدر ما كان منهم، ويثقون بما يحصل لهم من الشكر والإجلال والإعظام، راضين ولو بقليل من الثناء، قليلاً مقارنةً بما بذلوه من الجهد والمشقة في كشف خفايا العلم وإزالة ستائره.

Some applied themselves to obtain information which was not known before them, and left it to posterity; others commented upon the difficulties in the works left by their predecessors, and defined the best method (of study), or rendered the access (to science) easier or placed it more within reach; others again discovered mistakes in preceding works, and arranged that which was confused, or adjusted what was irregular, and corrected the faults of their fellow-labourers, without arrogance towards them, or taking pride in what they did themselves.

That fondness for science, by which God has distinguished the Imām al Mā'mun, the Commander of the Faithful (besides the caliphat which He has vouchsafed unto him by lawful succession, in the robe of which He has invested him, and with the honours of which He has adorned him), that affability and condescension which he shows to the learned, that promptitude with which he protects and supports them in the elucidation of obscurities and in the removal of difficulties—has encouraged me to compose a short work on Calculating by (the rules of) Completion and Reduction, confining it to what is easiest and most useful in arithmetic, such as men constantly require in cases of inheritance, legacies, partition, law-suits, and trade, and in all their dealings with one another, or where the measuring of lands, the digging of canals, geometrical computation, and other objects of various sorts and kinds are concerned—relying on the goodness of my intention therein, and hoping that the learned will reward it, by obtaining (for me) through their prayers the excellence of the Divine mercy: in requital of which, may the choicest blessings and the abundant bounty of God be theirs! My confidence rests with God, in this as in everything, and in Him I put my trust. He is the Lord of the Sublime Throne. May His blessing descend upon all the prophets and heavenly messengers!

وذلك الحب للعلم الذي ميَّز الله به الإمام المأمون أمير المؤمنين (مع ما منَّ به عليه من الخلافة بالإجماع، واللباس الذي ألبسه إياه، والزينة التي زيَّنه بها)، وذلك اللطف والتودد الذي يبديه للعلماء، وتلك السرعة التي يحميها وينصرها في توضيح المبهمات وإزالة الصعوبات قد حملني على أن أؤلف مختصراً في الجبر والمقابلة، أقتصر فيه على ما هو أيسر وأنفع في الحساب، مما يحتاج إليه الناس دائماً في مسائل الميراث والعطايا والقسمة والخصومات والتجارة وكل ما يتعاملون به فيما بينهم، أو في مساحة الأرضين وحفر الأنهار ومسائل الهندسة وغيرها من أنواع الأشياء المختلفة الأصناف.

Chapter 1

Introduction: On Numbers and the Six Cases

When I considered what people generally want in calculating, I found that it always is a number. I also observed that every number is composed of units, and that any number may be divided into units. Moreover, I found that every number, which may be expressed from one to ten, surpasses the preceding by one unit: afterwards the ten is doubled or tripled, just as before the units were: thus arise twenty, thirty, etc., until a hundred; then the hundred is doubled and tripled in the same manner as the units and the tens, up to a thousand; then the thousand can be thus repeated at any complex number; and so forth to the utmost limit of numeration.

لما نظرت فيما يحتاج إليه الناس في حساباتهم، وجدت أن ذلك كله عدد. ووجدت أيضاً أن كل عدد مؤلف من الوحدات، وأنه لا يخلو عدد من أن ينقسم إلى وحدات. وعلمت أن كل عدد يعدَّ من الواحد إلى العشرة يزيد على الذي قبله بواحد، ثم يضاعف العشر ويثلث، كما فعل بالوحدات، فيكون عشرون وثلاثون وهكذا إلى المائة، ثم يضاعف المائة ويثلثها كما فعل بالعشرات والوحدات إلى الألف، ثم يكرر الألف كذلك في كل عدد مركب، وهكذا إلى أقصى حدٍّ في العدِّ.

I observed that the numbers which are required in calculating by Completion and Reduction are of **three kinds**, namely, *roots*, *squares*, and *simple numbers* relative to neither root nor square.

The Three Kinds of Numbers

1. A **root** (*jadhar* / جذر) is any quantity which is to be multiplied by itself, consisting of units, or numbers ascending, or fractions descending.[a]

2. A **square** (*māl* / مال) is the whole amount of the root multiplied by itself.[b]

3. A **simple number** ($'adadmufrad$ / عدد مفرد) is any number which may be pronounced without reference to root or square. This is the known constant term.

[a] **[Ed.]** The Arabic term *jadhar* means "root." In modern notation, this is the unknown quantity x.
[b] **[Ed.]** The Arabic term *māl*, meaning "property" or "wealth," designates the square of the unknown. In modern notation: x^2.

فوجدت أن الأعداد التي يحتاج إليها في حساب الجبر والمقابلة ثلاثة أنواع: جذور، وأموال، وأعداد مفردة لا تتعلق بجذر ولا بمال. أما الجذر فهو كل مقدار يضرب في نفسه مؤلفاً من وحدات أو أعداد صاعدة أو كسور نازلة. وأما المال فهو جميع ما حصل من ضرب الجذر في نفسه. وأما العدد المفرد فهو كل عدد يذكر بغير نسبة إلى جذر ولا مال.

A number belonging to one of these three classes may be equal to a number of another class; you may say, for instance, "squares are equal to roots," or "squares are equal to numbers," or "roots are equal to numbers."

1.1 The Six Canonical Cases

From these three kinds, **six cases** arise—three simple and three compound. Al-Khwārizmī presents them in a well-defined order that would become standard for centuries of algebraic practice.

1.1.1 The Three Simple Cases

These cases do not require that the roots be halved (i.e., they can be solved by direct reduction):

Simple Cases (do not require halving)

Case I: Squares equal to Roots: $ax^2 = bx \quad \Longrightarrow \quad x = \dfrac{b}{a}$

Case II: Squares equal to Numbers: $ax^2 = c \quad \Longrightarrow \quad x = \sqrt{\dfrac{c}{a}}$

Case III: Roots equal to Numbers: $bx = c \quad \Longrightarrow \quad x = \dfrac{c}{b}$

هذه هي الحالات الستّ التي ذكرتها في مقدمة هذا الكتاب. ثلاث منها لا تحتاج إلى أن تنصف الجذور، وقد بيَّنت كيف تجاب.
وأما الثلاث الأخرى التي يجب فيها نصف الجذور، فأرى أن أبيّنها أيضاً مفصَّلة مع ذكل الشكل لكل حالة يبين سبب النصف.

Case I: Squares equal to Roots

Example: "A square is equal to five roots of the same." The root of the square is five, and the square is twenty-five, which is equal to five times its root.

Example: "One third of the square is equal to four roots." Then the whole square is equal to twelve roots; that is a hundred and forty-four; and its root is twelve.

Example: "Five squares are equal to ten roots." Then one square is equal to two roots; the root of the square is two, and its square is four.

$$\begin{gathered} \boxed{x^2 = 5x \quad \Rightarrow \quad x = 5, \quad x^2 = 25} \\ \boxed{\tfrac{1}{3}x^2 = 4x \quad \Rightarrow \quad x^2 = 12x = 144, \quad x = 12} \\ \boxed{5x^2 = 10x \quad \Rightarrow \quad x^2 = 2x = 4, \quad x = 2} \end{gathered} \tag{1.1}$$

Case II: Squares equal to Numbers

Example: "A square is equal to nine." Then this is a square, and its root is three.

Example: "Five squares are equal to eighty." Then one square is equal to one-fifth of eighty, which is sixteen.

Example: "The half of the square is equal to eighteen." Then the square is thirty-six, and its root is six.

$$\begin{gathered} \boxed{x^2 = 9 \quad \Rightarrow \quad x = 3} \\ \boxed{5x^2 = 80 \quad \Rightarrow \quad x^2 = 16, \quad x = 4} \\ \boxed{\tfrac{1}{2}x^2 = 18 \quad \Rightarrow \quad x^2 = 36, \quad x = 6} \end{gathered} \tag{1.2}$$

Case III: Roots equal to Numbers

Example: "One root equals three in number." Then the root is three, and its square nine.

Example: "Four roots are equal to twenty." Then one root is equal to five, and the square to be formed of it is twenty-five.

Example: "Half the root is equal to ten." Then the whole root is equal to twenty, and the square which is formed of it is four hundred.

$$\boxed{x = 3}$$
$$\boxed{4x = 20 \quad \Rightarrow \quad x = 5, \quad x^2 = 25} \tag{1.3}$$
$$\boxed{\tfrac{1}{2}x = 10 \quad \Rightarrow \quad x = 20, \quad x^2 = 400}$$

1.1.2 The Three Compound Cases

These cases require that the roots be halved (the method of *al-jabr*):

Compound Cases (require halving the roots)

(IV) **Squares and Roots equal to Numbers:** $ax^2 + bx = c$

(V) **Squares and Numbers equal to Roots:** $ax^2 + c = bx$

(VI) **Roots and Numbers equal to Squares:** $bx + c = ax^2$

Case IV: Squares and Roots equal to Numbers

Example: "One square, and ten roots of the same, amount to thirty-nine dirhems."

> **Solution:** You halve the number of the roots, which in the present instance yields five. This you multiply by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of the roots, which is five; the remainder is three. This is the root of the square which you sought for; the square itself is nine.

The Algorithm for Case IV

$$x^2 + bx = c \quad \Longrightarrow \quad x = \sqrt{\left(\frac{b}{2}\right)^2 + c} - \frac{b}{2} \tag{1.4}$$

Applied: $x^2 + 10x = 39 \quad \Rightarrow \quad x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$

مثال: مال وعشرة أصوله تساوي تسعة وثلاثين درهماً. يعني: ما هو المال الذي إذا أُضيف إليه عشرة من جذوره بلغ تسعة وثلاثين؟
الحل: نصف عدد الجذور، فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فنضيفه إلى تسعة وثلاثين فيكون تسعة وستين،
فتأخذ جذره فيكون ثمانية، فتنقص منه نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا هو جذر المال المطلوب، والمال نفسه تسعة.

The solution is the same when two squares or three, or more or less be specified; you reduce them to one single square, and in the same proportion you reduce also the roots and simple numbers which are connected therewith.

Example: "Two squares and ten roots are equal to forty-eight dirhems."

You must at first reduce the two squares to one; and you know that one square of the two is the moiety of both. Then reduce everything mentioned in the statement to its half: “a square and five roots of the same are equal to twenty-four dirhems.” Now halve the number of the roots; the moiety is two and a half. Multiply that by itself: the product is six and a quarter. Add this to twenty-four; the sum is thirty and a quarter. Take the root of this; it is five and a half. Subtract from this the moiety of the number of the roots, that is two and a half; the remainder is three. This is the root of the square, and the square itself is nine.

$$2x^2 + 10x = 48 \quad \Rightarrow \quad x^2 + 5x = 24 \quad \Rightarrow \quad x = \sqrt{6\tfrac{1}{4} + 24} - 2\tfrac{1}{2} = 5\tfrac{1}{2} - 2\tfrac{1}{2} = 3 \tag{1.5}$$

Case V: Squares and Numbers equal to Roots

Example: “A square and twenty-one in numbers are equal to ten roots of the same square.”

Solution: Halve the number of the roots; the moiety is five. Multiply this by itself; the product is twenty-five. Subtract from this the twenty-one which are connected with the square; the remainder is four. Extract its root; it is two. Subtract this from the moiety of the roots, which is five; the remainder is three. This is the root of the square which you required, and the square is nine. **Or** you may add the root to the moiety of the roots; the sum is seven; this is the root of the square which you sought for, and the square itself is forty-nine.

The Algorithm for Case V

$$x^2 + c = bx \quad \Longrightarrow \quad x = \frac{b}{2} \pm \sqrt{\left(\frac{b}{2}\right)^2 - c} \tag{1.6}$$

Applied: $x^2 + 21 = 10x \quad \Rightarrow \quad x = 5 \pm \sqrt{25 - 21} = 5 \pm 2 = 3$ or 7

مثال: مال وواحد وعشرون عدداً يساوي عشرة أصوله.
الحل: نصف عدد الجذور فيكون خمسة، ونضربه في نفسه فيكون خمسة وعشرين، فننقص منه الواحد والعشرين الملحقين بالمال، يبقى أربعة، فنستخرج جذرها فيكون اثنين، فننقصه من نصف عدد الجذور وهو خمسة، يبقى ثلاثة، وهذا جذر المال المطلوب، والمال تسعة. أو تضيف الجذر إلى نصف الجذور، فيكون سبعة، وهذا جذر المال المطلوب، والمال تسعة وأربعون.

When you meet with an instance which refers you to this case, try its solution by addition, and if that do not serve, then subtraction certainly will. For in this case both addition and subtraction may be employed, which will not answer in any other of the three cases in which the number of the roots must be halved.

Note: And know that, when in a question belonging to this case you have halved the number of the roots and multiplied the moiety by itself, if the product be **less** than the number of dirhems connected with the square, then the instance is **impossible**; but if the product be **equal** to the dirhems by themselves, then the root of the square is equal to the moiety of the roots alone, without either addition or subtraction.

Case VI: Roots and Numbers equal to Squares

Example: “Three roots and four of simple numbers are equal to a square.”

Solution: Halve the roots; the moiety is one and a half. Multiply this by itself; the product is two and a quarter. Add this to the four; the sum is six and a quarter. Extract its root; it is two and a half. Add this to the moiety of the roots, which was one and a half; the sum is four. This is the root of the square, and the square is sixteen.

The Algorithm for Case VI

$$bx + c = x^2 \quad \Longrightarrow \quad x = \frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + c} \tag{1.7}$$

Applied: $3x + 4 = x^2 \quad \Rightarrow \quad x = 1\frac{1}{2} + \sqrt{2\frac{1}{4} + 4} = 1\frac{1}{2} + 2\frac{1}{2} = 4$

مثال: ثلاثة أصول وأربعة أعداد يساوي مالاً.
الحل: نصف الجذور فيكون واحداً ونصفاً، ونضربه في نفسه فيكون اثنين وربعاً، فنضيفه إلى الأربعة فيكون ستة وربعاً، فنستخرج جذرها فيكون اثنين ونصفاً، فنضيفه إلى نصف الجذور وهو واحد ونصف فيكون أربعة، وهذا جذر المال، والمال ستة عشر.

Chapter 2

Geometrical Demonstrations

These are the six cases which I mentioned in the introduction to this book. They have now been explained. I have shown that three among them do not require that the roots be halved, and I have taught how they must be resolved. As for the other three, in which halving the roots is necessary, I think it expedient, more accurately, to explain them by separate chapters, in which a figure will be given for each case, to point out the reasons for halving.

2.1 Demonstration of Case IV: “A Square and Ten Roots equal to 39”

The figure to explain this is a **quadrate**, the sides of which are unknown. It represents the square, the root of which you wish to know. This is the figure AB, each side of which may be considered as one of its roots; and if you multiply one of these sides by any number, then the amount of that number may be looked upon as the number of the roots which are added to the square.

Each side of the quadrate represents the root of the square; and, as in the instance the roots were connected with the square, we may take one-fourth of ten, that is to say, two and a half, and combine it with each of the four sides of the figure. Thus with the original quadrate AB, four new parallelograms are combined, each having a side of the quadrate as its length, and the number of two and a half as its breadth; they are the parallelograms C, G, T, and K.

We have now a quadrate of equal, though unknown sides; but in each of the four corners of which a square piece of two and a half multiplied by two and a half is wanting. In order to compensate for this want and to complete the quadrate, we must add (to that which we have already) four times the square of two and a half, that is, twenty-five.

We know (by the statement) that the first figure, namely, the quadrate representing the square, together with the four parallelograms around it, which represent the ten roots, is equal to thirty-nine of numbers. If to this we add twenty-five, which is the equivalent of the four quadrates at the corners of the figure AB, by which the great figure DH is completed, then we know that this together makes sixty-four. One side of this great quadrate is its root, that is, eight. If we subtract twice a fourth of ten, that is five, from eight, as from the two extremities of the side of the great quadrate DH, then the remainder of such a side will be three, and that is the root of the square, or the side of the original figure AB.

Geometric Proof of Case IV

Given: $x^2 + 10x = 39$

$\boxed{\text{Square } AB}$	$=$	x^2
+ 4 parallelograms	$=$	$10x$
Subtotal	$=$	39
+ 4 corner squares $(2\frac{1}{2} \times 2\frac{1}{2})$	$=$	25
Total (Great Square DH)	$=$	64
Side of Great Square	$=$	$\sqrt{64} = 8$
$x = 8 - 2 \times 2\frac{1}{2} = 8 - 5 =$	$\boxed{3}$	

It must be observed, that we have halved the number of the roots, and added the product of the moiety multiplied by itself to the number thirty-nine, in order to complete the great figure in its four corners; because the fourth of any number multiplied by itself, and then by four, is equal to the product of the moiety of that number multiplied by itself.

2.2 Demonstration of Case V: "A Square and 21 equal to 10 Roots"

We represent the square by a quadrate AD, the length of whose side we do not know. To this we join a parallelogram, the breadth of which is equal to one of the sides of the quadrate AD, such as the side HN. This parallelogram is HB. The length of the two figures together is equal to the line HC. We know that its length is ten of numbers; for every quadrate has equal sides and angles, and one of its sides multiplied by a unit is the root of the quadrate, or multiplied by two it is twice the root of the same.

As it is stated, therefore, that a square and twenty-one of numbers are equal to ten roots, we may conclude that the length of the line HC is equal to ten of numbers, since the line CD represents the root of the square. We now divide the line CH into two equal parts at the point G: the line GC is then equal to HG.

We know that the quadrangle HB represents the twenty-one of numbers which were added to the quadrate. The whole quadrate MT was found to be equal to twenty-five. If we now subtract from this quadrate, MT, the quadrangles HT and MR, which are equal to twenty-one, there remains a small quadrate KR, which represents the difference between twenty-five and twenty-one. This is four; and its root, represented by the line RG, which is equal to GA, is two.

Geometric Proof of Case V

Given: $x^2 + 21 = 10x$

$$MT = \left(\frac{10}{2}\right)^2 = 25$$
$$HT + MR = HB = 21$$
$$KR = 25 - 21 = 4 \quad \Rightarrow \quad \sqrt{KR} = 2$$

$$x = GC - GA = 5 - 2 = 3 \quad \text{OR} \quad x = GC + GA = 5 + 2 = 7$$

If you subtract this number two from the line CG, which is the moiety of the roots, then the remainder is the line AC; that is to say, three, which is the root of the original square. But if you add the number two to the line CG, which is the moiety of the number of the roots, then the sum is seven, represented by the line CR, which is the root to a larger square. However, if you add twenty-one to this square, then the sum will likewise be equal to ten roots of the same square.

2.3 Demonstration of Case VI: "3 Roots and 4 equal to a Square"

Let the square be represented by a quadrangle, the sides of which are unknown to us, though they are equal among themselves, as also the angles. This is the quadrate AD, which comprises the three roots and the four of numbers mentioned in this instance.

In every quadrate one of its sides, multiplied by a unit, is its root. We now cut off the quadrangle HD from the quadrate AD, and take one of its sides HC for three, which is the number of the roots. The same is equal to RD. It follows, then, that the quadrangle HB represents the four of numbers which are added to the roots. Now we halve the side CH, which is equal to three roots, at the point G; from this division we construct the square HT, which is the product of half the roots (or one and a half) multiplied by themselves, that is to say, two and a quarter.

Geometric Proof of Case VI

Given: $3x + 4 = x^2$

$$\text{Square } HT = \left(\tfrac{3}{2}\right)^2 = 2\tfrac{1}{4}$$
$$AN + KL = AR = 4 \text{ (the added numbers)}$$
$$GM = HT + (AN + KL) = 2\tfrac{1}{4} + 4 = 6\tfrac{1}{4}$$
$$AG = \sqrt{GM} = 2\tfrac{1}{2}$$
$$x = AG + GC = 2\tfrac{1}{2} + 1\tfrac{1}{2} = \boxed{4}$$

Chapter 3

On Multiplication of Algebraic Expressions

I SHALL NOW TEACH YOU how to multiply the unknown numbers, that is to say, the roots, one by the other, if they stand alone, or if numbers are added to them, or if numbers are subtracted from them, or if they are subtracted from numbers; also how to add them one to the other, or how to subtract one from the other.

Whenever one number is to be multiplied by another, the one must be repeated as many times as the other contains units.

If there are greater numbers combined with units to be added to or subtracted from them, then **four multiplications** are necessary; namely, the greater numbers by the greater numbers, the greater numbers by the units, the units by the greater numbers, and the units by the units.

If the units, combined with the greater numbers, are both positive, then the last multiplication is positive; if they are both negative, then the fourth multiplication is likewise positive. But if one of them is positive, and one negative, then the fourth multiplication is negative.

The Fourfold Multiplication Rule

To multiply $(a + x)$ by $(b + y)$:

1. Greater by greater: $a \times b$
2. Greater by units: $a \times y$
3. Units by greater: $x \times b$
4. Units by units: $x \times y$

Sign rule for the fourth multiplication: $(+x)(+y) = +xy \quad (-x)(-y) = +xy \quad (+x)(-y) = -xy \quad (-x)(+y) = -xy$

3.1 Worked Examples

Example 1: "Ten and one to be multiplied by ten and two."

Ten times ten is a hundred; once ten is ten positive; twice ten is twenty positive, and once two is two positive; this altogether makes a hundred and thirty-two.

$(10 + 1)(10 + 2) = 100 + 10 + 20 + 2 = 132$

Example 2: "Ten less one, to be multiplied by ten less one."

Then ten times ten is a hundred; the negative one by ten is ten negative; the other negative one by ten is likewise ten negative, so that it becomes eighty: but the negative one by the negative one is one positive, and this makes the result eighty-one.

$(10-1)(10-1) = 100 - 10 - 10 + 1 = 81$

Example 3: "Ten and thing to be multiplied by ten and thing"

Then ten times ten is a hundred, and ten times thing is ten things; and again, ten times thing is ten things; and thing multiplied by thing is a square positive, so that the whole product is a hundred dirhems and twenty things and one positive square.

$(10+x)(10+x) = 100 + 10x + 10x + x^2 = 100 + 20x + x^2$

Example 4: "Ten minus thing to be multiplied by ten minus thing"

Then ten times ten is a hundred; and minus thing by ten is minus ten things; and again, minus thing by ten is minus ten things. But minus thing multiplied by minus thing is a positive square. The product is therefore a hundred and a square, minus twenty things.

$(10-x)(10-x) = 100 - 10x - 10x + x^2 = 100 - 20x + x^2$

Example 5: "Ten minus thing to be multiplied by ten and thing"

Then you say, ten times ten is a hundred; and minus thing by ten is ten things negative; and thing by ten is ten things positive; and minus thing by thing is a square positive; therefore, the product is a hundred dirhems, minus a square.

$(10-x)(10+x) = 100 - 10x + 10x - x^2 = 100 - x^2$

Chapter 4

On Addition and Subtraction

4.1 Operations on Radical Expressions

Know that the root of two hundred minus ten, added to twenty minus the root of two hundred, is just ten.

$(\sqrt{200} - 10) + (20 - \sqrt{200}) = 10$

The root of two hundred, minus ten, subtracted from twenty minus the root of two hundred, is thirty minus twice the root of two hundred; twice the root of two hundred is equal to the root of eight hundred.

$(20 - \sqrt{200}) - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800}$

A hundred and a square minus twenty roots, added to fifty and ten roots minus two squares, is a hundred and fifty, minus a square and minus ten roots.

$(100 + x^2 - 20x) + (50 + 10x - 2x^2) = 150 - x^2 - 10x$

A hundred and a square, minus twenty roots, diminished by fifty and ten roots minus two squares, is fifty dirhems and three squares minus thirty roots.

$(100 + x^2 - 20x) - (50 + 10x - 2x^2) = 50 + 3x^2 - 30x$

4.2 On Doubling, Tripling, and Halving Roots

If you require to double the root of any known or unknown square, (the meaning of its duplication being that you multiply it by two) then it will suffice to multiply two by two, and then by the square; the root of the product is equal to twice the root of the original square.

$2\sqrt{S} = \sqrt{4S}$

If you require to take it thrice, you multiply three by three, and then by the square; the root of the product is thrice the root of the original square.

$3\sqrt{S} = \sqrt{9S}$

If you require to find the moiety of the root of the square, you need only multiply a half by a half, which is a quarter; and then this by the square: the root of the product will be half the root of the first square.

$\frac{1}{2}\sqrt{S} = \sqrt{\frac{1}{4}S}$

Chapter 5

On Division

If you will divide the root of nine by the root of four, you begin with dividing nine by four, which gives two and a quarter: the root of this is the number which you require—it is one and a half.

$$\frac{\sqrt{9}}{\sqrt{4}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\tfrac{1}{2}$$

If you will divide the root of four by the root of nine, you divide four by nine; it is four-ninths of the unit; the root of this is two divided by three; namely, two-thirds of the unit.

$$\frac{\sqrt{4}}{\sqrt{9}} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

If you wish to multiply the root of nine by the root of four, multiply nine by four; this gives thirty-six; take its root, it is six; this is the root of nine, multiplied by the root of four.

$$\sqrt{9} \times \sqrt{4} = \sqrt{9 \times 4} = \sqrt{36} = 6$$

If you wish to multiply twice the root of nine by thrice the root of four, then take twice the root of nine, according to the rule above given, so that you may know the root of what square it is. You do the same with respect to the three roots of four in order to know what must be the square of such a root. You then multiply these two squares, the one by the other, and the root of the product is equal to twice the root of nine, multiplied by thrice the root of four.

$$(2\sqrt{9}) \times (3\sqrt{4}) = \sqrt{36 \times 36} = 36$$

Chapter 6

The Six Illustrative Problems

Before the chapters on computation and the several species thereof, I shall now introduce six problems, as instances of the six cases treated of in the beginning of this work. I have shown that three among these cases, in order to be solved, do not require that the roots be halved, and I have also mentioned that the calculating by completion and reduction must always necessarily lead you to one of these cases. I now subjoin these problems, which will serve to bring the subject nearer to the understanding, to render its comprehension easier, and to make the arguments more perspicuous.

First Problem

Statement: "I have divided ten into two portions; I have multiplied the one of the two portions by the other; after this I have multiplied the one of the two by itself, and the product of the multiplication by itself is four times as much as that of one of the portions by the other."

Solution: Suppose one of the portions to be x, and the other $10 - x$. You multiply x by $10 - x$; it is $10x - x^2$. Then multiply it by four, because the instance states "four times as much." The result will be $40x - 4x^2$. After this you multiply x by x, that is to say, one of the portions by itself. This is a square, which is equal to $40x - 4x^2$. Reduce it now by the four squares, and add them to the one square. Then the equation is: forty things are equal to five squares; and one square will be equal to eight roots, that is, sixty-four; the root of this is eight.

$$x^2 = 4x(10-x) = 40x - 4x^2 \quad \Rightarrow \quad 5x^2 = 40x \quad \Rightarrow \quad x^2 = 8x = 64 \quad \Rightarrow \quad x = 8 \tag{6.1}$$

The portions are: $x = 8$ and $10 - x = 2$.

This question leads you to one of the six cases, namely, that of "squares equal to roots."

Second Problem

Statement: "I have divided ten into two portions: I have multiplied each of the parts by itself, and afterwards ten by itself: the product of ten by itself is equal to one of the two parts multiplied by itself, and afterwards by two and seven-ninths."

$$100 = x^2 \times 2\tfrac{7}{9} = x^2 \times \tfrac{25}{9} \quad \Rightarrow \quad x^2 = \tfrac{900}{25} = 36 \quad \Rightarrow \quad x = 6 \tag{6.2}$$

The portions are: $x = 6$ and $10 - x = 4$.

Third Problem

Statement: "I have divided ten into two parts. I have afterwards divided the one by the other, and the quotient was four."

$$\frac{10-x}{x} = 4 \quad \Rightarrow \quad 10 - x = 4x \quad \Rightarrow \quad 10 = 5x \quad \Rightarrow \quad x = 2 \tag{6.3}$$

The portions are: $x = 2$ and $10 - x = 8$.

Fourth Problem

Statement: "I have multiplied one-third of thing and one dirhem by one-fourth of thing and one dirhem, and the product was twenty."

$$\left(\frac{x}{3}+1\right)\left(\frac{x}{4}+1\right) = 20 \quad \Rightarrow \quad \frac{x^2}{12} + \frac{7x}{12} + 1 = 20 \quad \Rightarrow \quad x^2 + 7x = 228 \tag{6.4}$$

Halve the roots: $3\frac{1}{2}$. Square: $12\frac{1}{4}$. Add to 228: $240\frac{1}{4}$. Root: $15\frac{1}{2}$. Subtract $3\frac{1}{2}$: $x = 12$.

Fifth Problem

Statement: "I have divided ten into two parts; and having multiplied each part by itself, I have added them together; and the sum was fifty-eight dirhems."

$$x^2 + (10-x)^2 = 58 \quad \Rightarrow \quad 2x^2 - 20x + 100 = 58 \quad \Rightarrow \quad x^2 + 28 = 11x \tag{6.5}$$

$x = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = 5\frac{1}{2} \pm 2\frac{1}{2}$. The portions are: $x = 3$ or $x = 7$.

Sixth Problem

Statement: "I have multiplied one-third of a root by one-fourth of a root, and the product is equal to the root and twenty-four dirhems."

$$\frac{x}{3} \cdot \frac{x}{4} = x + 24 \quad \Rightarrow \quad \frac{x^2}{12} = x + 24 \quad \Rightarrow \quad x^2 = 12x + 288 \tag{6.6}$$

$x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$

Chapter 7

Various Questions

Further Illustrative Problems

Problem 1: "I have divided ten into two parts, and multiplying one of these by the other, the result was twenty-one."

$$x(10-x)=21 \;\Rightarrow\; 10x-x^2=21 \;\Rightarrow\; x^2+21=10x \; x=5\pm\sqrt{25-21}=5\pm 2=3 \text{ or } 7$$

Problem 2: "I have divided ten into two parts; having multiplied each part by itself, I have subtracted the smaller from the greater, and the remainder was forty."

$$(10-x)^2-x^2=40 \;\Rightarrow\; 100-20x=40 \;\Rightarrow\; x=3$$

Problem 3: "There is a square, two-thirds of one-fifth of which are equal to one-seventh of its root."

$$\tfrac{2}{3}\cdot\tfrac{1}{5}x^2=\tfrac{1}{7}x \;\Rightarrow\; \tfrac{2}{15}x^2=\tfrac{1}{7}x \;\Rightarrow\; x=\tfrac{15}{14}=1\tfrac{1}{14}$$

Problem 4: "To find a square-root which, when multiplied by four times itself, amounts to twenty."

$$x\cdot 4x=20 \;\Rightarrow\; 4x^2=20 \;\Rightarrow\; x^2=5 \;\Rightarrow\; x=\sqrt{5}$$

Problem 5: "A square, four of the roots of which multiplied by five of its roots produce twice the square, with a surplus of thirty-six dirhems."

$$4x\cdot 5x=2x^2+36 \;\Rightarrow\; 20x^2=2x^2+36 \;\Rightarrow\; 18x^2=36 \;\Rightarrow\; x^2=2$$

Problem 6: "To find a square, of which if one-third be added to three dirhems, and the sum be subtracted from the square, the remainder multiplied by itself restores the square."

$$\left(x-\tfrac{x}{3}-3\right)^2=x \;\Rightarrow\; \left(\tfrac{2x}{3}-3\right)^2=x \;\Rightarrow\; \tfrac{4x^2}{9}+9-4x=x \text{ Complete the square: } x^2+20\tfrac{1}{4}=11\tfrac{1}{4}x \;\Rightarrow\; x=9 \text{ or } 2\tfrac{1}{4}$$

Chapter 8

On Mercantile Transactions

YOU KNOW THAT all mercantile transactions of people, such as buying and selling, exchange and hire, comprehend always **two notions** and **four numbers**, which are stated by the enquirer; namely, *measure* and *price*, and *quantity* and *sum*.

The number which expresses the measure is inversely proportionate to the number which expresses the sum, and the number of the price inversely proportionate to that of the quantity. Three of these four numbers are always known, one is unknown, and this is implied when the person inquiring says *how much?* and it is the object of the question.

The Rule of Four Numbers

Measure	Price
a (known)	b (known)
c (known or unknown)	d (known or unknown)

$$a : c :: d : b \quad \text{or equivalently} \quad a \times b = c \times d$$

$$\text{Unknown} = \frac{\text{Product of the two proportionate numbers}}{\text{The third known number}}$$

Examples

Example 1: "Ten for six, how much for four?"

Here ten is the measure; six is the price; "how much" implies the unknown quantity; and four is the number of the sum. The measure (10) is inversely proportionate to the sum (4).

$$\text{Quantity} = \frac{10 \times 4}{6} = \frac{40}{6} = 6\tfrac{2}{3}$$

Example 2: "Ten for eight, what must be the sum for four?"

$$\text{Sum} = \frac{8 \times 4}{10} = \frac{32}{10} = 3\tfrac{1}{5}$$

Example 3: "A workman receives a pay of ten dirhems per month; how much must be his pay for six days?"

Six days are one-fifth of the month; and his portion of the dirhems must be proportionate to the portion of the month.

$$\text{Pay} = \frac{10 \times 6}{30} = 2 \text{ dirhems}$$

المعاملات التجارية: اعلم أن جميع معاملات الناس من بيع وشراء ومصارفة وإجارة، فإنها لا تخلو من قدرين وأربعة أعداد يسأل عنها السائل، وهي المقدار والثمن والكيفية والجميع. عدد المقدار يناسب عدد الجميع، وعدد الثمن يناسب عدد الكيفية. ثلاثة من هذه

الأربعة دائماً معلومة، وواحد مجهول، وهو الذي يتضمنه قوله «كم؟» وهو المطلوب.

Chapter 9

Mensuration

KNOW THAT the meaning of the expression "one by one" is mensuration: one yard (in length) by one yard (in breadth) being understood.

Every quadrangle of equal sides and angles, which has one yard for every side, has also one for its area. Has such a quadrangle two yards for its side, then the area of the quadrangle is four times the area of a quadrangle, the side of which is one yard.

Triangles

If you multiply the height of any equilateral triangle by the moiety of the basis upon which the line marking the height stands perpendicularly, the product gives the area of that triangle.

$$\text{Area of triangle} = \text{height} \times \frac{\text{basis}}{2} \tag{9.1}$$

Circles

In any circle, the product of its diameter, multiplied by three and one-seventh, will be equal to the periphery.

$$C = \pi d \approx \frac{22}{7} \times d \tag{9.2}$$

The area of any circle will be found by multiplying the moiety of the circumference by the moiety of the diameter.

$$A = \frac{C}{2} \times \frac{d}{2} = \frac{Cd}{4} \tag{9.3}$$

If you multiply the diameter of any circle by itself, and subtract from the product one-seventh and half one-seventh of the same, then the remainder is equal to the area of the circle.

$$A = d^2 - \frac{d^2}{7} - \frac{d^2}{14} = d^2 \times \left(1 - \frac{3}{14}\right) = d^2 \times \frac{11}{14} \approx \frac{\pi}{4} d^2 \tag{9.4}$$

Note: This gives $\pi \approx \frac{22}{7}$, the well-known approximation.

Volumes

The bulk of a quadrangular body will be found by multiplying the length by the breadth, and then by the height.

Cones and pyramids, such as triangular or quadrangular ones, are computed by multiplying one-third of the area of the basis by the height.

$$V_{\text{pyramid}} = \frac{1}{3} \times (\text{area of base}) \times \text{height} \tag{9.5}$$

The Right Triangle (Pythagorean Theorem)

Observe, that in every rectangular triangle the two short sides, each multiplied by itself and the products added together, equal the product of the long side multiplied by itself.

The Pythagorean Theorem

$$a^2 + b^2 = c^2 \tag{9.6}$$

Where a and b are the two shorter sides (catheti) and c is the hypotenuse (long side).

Al-Khwārizmī provides a beautiful geometric proof of this theorem using a quadrate divided into four equal smaller quadrates and eight equal triangles, demonstrating that the sum of the squares on the two shorter sides equals the square on the longest side.

Chapter 10

On Legacies and the Laws of Inheritance

The final portion of this work treats of the Islamic laws of inheritance (*al-farā'iḍ*). This subject, while seemingly belonging to jurisprudence rather than mathematics, requires the full apparatus of algebra for its solution when legacies are superimposed upon the statutory shares.

On Capital and Money Lent

Problem: "A man dies, leaving two sons behind him, and bequeathing one-third of his capital to a stranger. He leaves ten dirhems of property and a claim of ten dirhems upon one of the sons."

Solution: You call the sum which is taken out of the debt x (thing). Add this to the capital, which is ten dirhems. The sum is $10 + x$. Subtract one-third of this, since he has bequeathed one-third of his property: $3\frac{1}{3} + \frac{1}{3}x$. The remainder is $6\frac{2}{3} + \frac{2}{3}x$. Divide this between the two sons: each receives $3\frac{1}{3} + \frac{1}{3}x$. This is equal to the thing which was sought for.

$$3\tfrac{1}{3} + \tfrac{1}{3}x = x \quad \Rightarrow \quad 3\tfrac{1}{3} = \tfrac{2}{3}x \quad \Rightarrow \quad x = 5$$

The stranger receives 5; and each son receives 5 (the indebted son retains his full debt as a set-off against his share).

On Another Species of Legacy

Problem: "A man dies, leaving his mother, his wife, and two brothers and two sisters by the same father and mother with himself; and he bequeaths to a stranger one-ninth of his capital."

Solution: You constitute their shares by taking them out of **forty-eight parts**. You know that if you take one-ninth from any capital, eight-ninths of it will remain. Add now to the eight-ninths one-eighth of the same, and to the forty-eight also one-eighth of them, namely, six, in order to complete your capital. This gives fifty-four. The person to whom one-ninth is bequeathed receives six out of this. The remaining forty-eight will be distributed among the heirs, proportionably to their legal shares.

Heir	Share (out of 48)
Mother	8
Wife	6
Each brother	14
Each sister	7
Total	$8 + 6 + 2(14) + 2(7) = 56$ parts adjusted

Complex Legacy Problems

Problem: "A woman dies, leaving her husband, a son, and her mother. She bequeaths to a person two-fifths, and to another one-fourth of her capital. She imposes the two legacies together on her son, and on her mother one moiety (of the mother's share of the residue); on her husband she imposes nothing but one-third (which he must contribute, according to the law)."

Solution: You constitute the shares of the heritage by taking them out of twelve parts: the son receives seven of them, the husband three, and the mother two parts. You know that the husband must give up one-third of his share; accordingly he retains twice as much as that which is detracted from his share for the legacy. As he has three parts in hand, one of these falls to the legacy, and the remaining two parts he retains for himself.

Heir	**Original Share**	**Disposition**
Son	7/12	Charged with $\frac{2}{5} + \frac{1}{4} = \frac{13}{20}$
Husband	3/12 = 1/4	Contributes 1/3 of share = 1/12
Mother	2/12 = 1/6	Charged with 1/2 of share = 1/12

Total contributed $= \frac{13}{20} \times \frac{7}{12} + \frac{1}{12} + \frac{1}{12} =$ computed proportionally

Appendix A

Summary of the Six Cases

For reference, the six canonical cases of algebra are summarized here in modern notation:

Case	Arabic Name	Equation	Solution
I	مال يساوي أصولاً	$ax^2 = bx$	$x = b/a$
II	مال يساوي عدداً	$ax^2 = c$	$x = \sqrt{c/a}$
III	أصول تساوي عدداً	$bx = c$	$x = c/b$
IV	مال وأصول يساوي عدداً	$ax^2 + bx = c$	$x = \sqrt{(b/2a)^2 + c/a} - b/2a$
V	مال وعدد يساوي أصولاً	$ax^2 + c = bx$	$x = \frac{b}{2a} \pm \sqrt{(b/2a)^2 - c/a}$
VI	أصول وعدد يساوي مالاً	$bx + c = ax^2$	$x = \frac{b}{2a} + \sqrt{(b/2a)^2 + c/a}$

Historical Note: These six cases exhaust all possibilities for quadratic equations with *positive* coefficients. The restriction to positive coefficients was necessitated by the geometric interpretation: al-Khwārizmī understood all terms as geometric magnitudes (areas and lengths), which cannot be negative. The full treatment of all quadratic equations, including negative coefficients, would have to wait until the Renaissance.

Appendix B

Glossary of Arabic Terms

Arabic	**Transliteration**	**Meaning**
الجبر	*al-jabr*	Completion, restoration (adding terms to both sides)
المقابلة	*al-muqābala*	Balancing, reduction (cancelling like terms)
ال جذر	*al-jadhr*	Root (the unknown quantity, x)
المال	*al-māl*	Square of the unknown (x^2)
العدد المفرد	*al-'adad al-mufrad*	Simple number (constant term)
الشيء	*al-shay'*	Thing (another term for the unknown)
الدرهم	*al-dirham*	Dirhem (unit of currency/measure)
الوصية	*al-waṣiyya*	Legacy, bequest
الفرائض	*al-farā'iḍ*	Obligatory shares (Islamic inheritance law)
مساحة	*misāḥa*	Area, mensuration
ضرب	*ḍarb*	Multiplication
قسمة	*qisma*	Division
جمع	*jam'*	Addition
طرح	*tarḥ*	Subtraction
جذر	*jadhr*	Root, square root

Bibliographic Note

Manuscript Sources:

The principal manuscript of al-Khwārizmī's *Kitāb al-Mukhtaṣar fī Ḥisāb al-Jabr wa-l-Muqābala* is preserved in the Bodleian Library at Oxford (MS Hunt. 214, fols. 1–123), copied in 743 AH / 1342 CE. It forms part of the codex designated CMXXVIII, which also contains three other treatises on arithmetic and algebra.

Editions and Translations:

[1] Frederic Rosen, *The Algebra of Mohammed ben Musa*, London: Oriental Translation Fund, 1831. (Arabic text with English translation)

[2] Louis Charles Karpinski, *Robert of Chester's Latin Translation of the Algebra of Al-Khowarizmi*, New York: Macmillan, 1915. (Latin text with English version)

[3] 'Alī Muṣṭafā Musharrafa and Muḥammad Mursī Aḥmad (eds.), *Kitāb al-Jabr wa-l-Muqābala*, Cairo: Egyptian University / Maṭba'at Fatḥ Allāh Ilyās Nūrī, 1359 AH / 1939 CE. (Critical Arabic edition, the source of the present

Further Reading:

- J. Ruska, "Zur
- J. Ruska, "Zur ältesten arabischen Algebra und Rechenkunst," *Sitzungsberichte der Heidelberger Akademie der Wissenschaften*, Philosophisch-historische Klasse, Jahrgang 1917, 8. Abhandlung.
- H. Suter, "Die Mathematiker und Astronomen der Araber und ihre Werke," *Abhandlungen zur Geschichte der mathematischen Wissenschaften*, 10. Heft, Leipzig, 1900.
- R. Rashed (ed.), *Al-Khwārizmī: Le commencement de l'algèbre*, Paris: Blanchard, 2007. (Critical edition with French translation)
- B. van Dalen et al., "Al-Khwārizmī's Astronomical Tables in a Tenth-Century Context: The *Kitāb al-Majisṭī* (*Codex Studii*) and the Hindu-Arabic Numerals in the West," *SCIAMVS*, 18 (2017), pp. 49–93.

تمّ بحمد الله